

Skills Summary

Multiplying Fractions and Mixed Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — Multiply fractions (and whole numbers)

Rule: $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$ — multiply straight across. No common denominator is needed.

- A whole number n is $\frac{n}{1}$, so the same rule covers it.
- Cancel a common factor from any top with any bottom *before* you multiply — the numbers stay small and the answer lands simplified.
- “of” means $\times$: $\frac{2}{3}$ of a cup is $\frac{2}{3} \times 1$ cup.

Example: $\frac{4}{9} \times \frac{3}{8}$

Step 1. Two fractions multiplied: multiply across, but cancel first so nothing needs reducing at the end.

Step 2. 3 divides into 9 and 4 divides into 8: $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6} \Rightarrow \frac{1}{6}$

Try it: $6 \times \frac{2}{9}$

$\frac{8}{4}$ check

Skill 2 — Multiply mixed numbers

Rule: Convert first, multiply second: $a\frac{b}{c} = \frac{a \times c + b}{c}$.

- Never multiply the whole parts and the fraction parts separately.
- Estimate to check: $2\frac{3}{4} \times 1\frac{1}{2}$ must be a bit more than 3×1 , so an answer near 4 is believable.

Example: $1\frac{2}{3} \times 2\frac{1}{4}$

Step 1. Mixed numbers cannot be multiplied part by part, so rewrite both as improper fractions and then use the across rule.

Step 2. $1\frac{2}{3} = \frac{5}{3}$ and $2\frac{1}{4} = \frac{9}{4}$; cancel 3 into 9: $\frac{5}{1} \times \frac{3}{4} = \frac{15}{4} \Rightarrow \frac{15}{4}$

Try it: $3\frac{1}{2} \times \frac{4}{7}$

2 check

Skill 3 — Find a missing factor

Rule: If $\frac{a}{b} \times x = p$, then $x = p \div \frac{a}{b} = p \times \frac{b}{a}$ — divide by a fraction by multiplying by its **reciprocal** (flip it).

Always check by multiplying your answer back into the original sentence.

Example: Solve $\frac{2}{5} \times x = \frac{6}{25}$

Step 1. One factor and the product are known, so undo the multiplication — divide the product by the known factor.

Step 2. $x = \frac{6}{25} \times \frac{5}{2} = \frac{3}{5}$, and $\frac{2}{5} \times \frac{3}{5} = \frac{6}{25} \checkmark \Rightarrow x = \frac{3}{5}$

Try it: Solve $x \times 1\frac{1}{2} = \frac{9}{4}$

check: $\frac{3}{2} \times 1\frac{1}{2} = \frac{9}{4}$